Comparative Evaluation of ORB-Based Bag-of-Visual-Words, Convolutional Neural Networks, and Transfer Learning for Herbal Plant Recognition

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Abstract

Automatic recognition of herbal plant species is an important computer vision task with applications in biodiversity conservation, agriculture, and traditional medicine. However, accurate plant classification remains challenging due to variations in lighting conditions, viewing angles, image quality, and background complexity. This paper presents a comparative study of traditional computer vision and deep learning approaches for herbal plant classification using a dataset constructed from 21 video sequences collected under diverse environmental conditions. A total of 425 images representing 14 plant classes were extracted and preprocessed through vegetation-based filtering, blur detection, image normalization, and data augmentation. Three classification strategies were investigated: (1) a traditional machine learning pipeline based on ORB feature extraction and Bag-of-Visual-Words representation, (2) a custom Convolutional Neural Network (CNN), and (3) transfer learning using MobileNetV2 pre-trained on ImageNet. Experimental results show that the traditional ORB-BoVW-SVM approach achieved a validation accuracy of 47.92% and a test accuracy of 36.46%, while the custom CNN achieved 22.40% test accuracy. In contrast, the transfer learning model significantly outperformed both approaches, achieving a test accuracy of 93.23%. These findings demonstrate the effectiveness of transfer learning for plant recognition tasks with limited training data and highlight the advantages of leveraging pre-trained deep feature representations over handcrafted features and shallow neural networks. The proposed framework provides a reproducible and effective solution for automated herbal plant identification in real-world environments.

1. Introduction

The automatic identification of plant species has become an important research area in computer vision due to its applications in agriculture, biodiversity conservation, environmental monitoring, and traditional medicine. Accurate plant recognition systems can assist researchers, farmers, and healthcare practitioners in identifying species efficiently without requiring extensive botanical expertise. With the increasing availability of digital imaging devices and machine learning techniques, image-based plant classification has emerged as a practical alternative to manual identification.

Despite recent advances, plant classification remains a challenging task. Images of plants often exhibit substantial variations in illumination, viewing angles, scale, background clutter, and image quality. These challenges are further amplified when data are collected in uncontrolled outdoor environments. Traditional computer vision approaches rely on handcrafted features designed to capture local image characteristics, but such methods frequently struggle to represent complex visual patterns and semantic information. In contrast, deep learning approaches have demonstrated remarkable success in learning hierarchical feature representations directly from image data, often achieving superior performance across a wide range of visual recognition tasks.

This project investigates the problem of herbal plant classification using a dataset constructed from 21 video recordings captured under diverse environmental and lighting conditions. Frames extracted from the videos were processed through a preprocessing pipeline consisting of vegetation filtering, blur detection, image normalization, and data augmentation. The resulting dataset contains 425 images distributed across 14 herbal plant classes.

To evaluate the effectiveness of different recognition strategies, three approaches were studied. First, a traditional machine learning pipeline was implemented using ORB feature extraction and a Bag-of-Visual-Words (BoVW) representation, followed by classification using Support Vector Machines (SVM), Random Forests, Logistic Regression, and Gradient Boosting. Second, a custom Convolutional Neural Network (CNN) was trained directly on the image data. Finally, transfer learning was employed using a MobileNetV2 model pre-trained on ImageNet to leverage rich feature representations learned from large-scale image datasets.

Experimental results reveal substantial differences among the evaluated methods. While the best traditional approach achieved moderate classification performance, the custom CNN struggled to generalize effectively on the relatively small dataset. The transfer learning model achieved the highest accuracy, demonstrating the effectiveness of pre-trained deep neural networks for herbal plant recognition under real-world conditions.

The main contributions of this work are summarized as follows:

- Construction of a herbal plant image dataset from video sequences collected under diverse environmental conditions.
- Development of a preprocessing pipeline incorporating vegetation filtering, blur detection, image normalization, and data augmentation.
- Comparative evaluation of traditional feature-based methods, a custom CNN, and transfer learning for herbal plant classification.
- Demonstration that transfer learning with MobileNetV2 substantially outperforms handcrafted feature-based approaches and CNN training from scratch on a limited dataset.

The remainder of this paper is organized as follows. Section 2 reviews related work on plant classification and image recognition techniques. Section 3 describes the proposed methodology, including dataset preparation, feature extraction, and model development. Section 4 presents the experimental setup and results. Finally, Section 5 concludes the paper and discusses future research directions.

2. Related Work

2.1. Plant Species Classification

Automated plant species identification has become an important application of computer vision due to its relevance in agriculture, biodiversity conservation, and traditional medicine. Image-based plant recognition systems aim to distinguish plant species using visual characteristics captured from digital images. However, variations in lighting conditions, viewing angles, and background complexity make this task challenging, particularly when images are collected in natural environments.

2.2. Feature-Based Image Classification

Traditional image classification methods rely on handcrafted feature descriptors to represent visual information. In this work, Oriented FAST and Rotated BRIEF (ORB) features were used to detect keypoints and generate descriptors that are robust to rotation and scale variations [3]. ORB provides an efficient alternative for extracting local image features and has been widely applied in computer vision tasks.

To obtain a fixed-length image representation, the extracted ORB descriptors were converted into a Bag-of-Visual-Words (BoVW) representation using K-Means clustering [5]. The resulting visual word histograms were then classified using machine learning algorithms including Support Vector Machines (SVMs), Random Forests, Logistic Regression, and Gradient Boosting. Although BoVW-based approaches can effectively summarize local image features, they do not preserve the spatial relationships between keypoints within an image.

2.3. Convolutional Neural Networks for Image Recognition

Convolutional Neural Networks (CNNs) automatically learn hierarchical feature representations directly from image pixels, eliminating the need for handcrafted descriptors. By learning both low-level and high-level visual patterns, CNNs have become a widely used approach for image classification tasks. In this work, a custom CNN architecture was trained from scratch to classify herbal plant images and to evaluate the effectiveness of learned features compared with traditional feature-based methods.

2.4. Transfer Learning for Small Datasets

Transfer learning is commonly used when the available training data are limited. Instead of training a deep network from scratch, a model pre-trained on a large dataset can be adapted to a new classification task. In this study, MobileNetV2 [4] pre-trained on ImageNet was used as the backbone network. MobileNetV2 employs depthwise separable convolutions, making it computationally efficient while maintaining strong classification performance.

In this work, we compare three approaches for herbal plant classification: a traditional ORB-BoVW machine learning pipeline, a custom CNN trained from scratch, and a MobileNetV2 transfer learning model. This comparison provides insights into the effectiveness of handcrafted features and deep learning methods for plant recognition under real-world environmental conditions.

3. Methodology

This section describes the dataset construction process, image preprocessing pipeline, feature extraction techniques, and classification models used for herbal plant recognition.

3.1. Dataset Construction

The dataset used in this study was constructed from 21 video clips containing different herbal plant species captured under diverse environmental and lighting conditions. The videos were processed frame-by-frame to extract representative images for classification. Since consecutive video

Table 1. Dataset statistics showing the number of preprocessed plant images per pseudo-label class (A through N).

Class	Image count	Class	Image count
A	11	H	30
B	43	I	9
C	27	J	16
D	37	K	38
E	33	L	40
F	34	M	47
G	29	N	31
Total			**425**

frames often contain redundant information, only frames satisfying predefined quality criteria were retained.

After preprocessing and filtering, a total of 425 images were obtained and organized into 14 plant classes labeled from A to N. To ensure reproducibility and unbiased evaluation, the dataset was divided into training, validation, and testing subsets using stratified sampling.

The final dataset split consisted of:

- Training set: 297 images (70%)
- Validation set: 64 images (15%)
- Test set: 64 images (15%)

3.2. Image Preprocessing

The raw video frames exhibited considerable variability in image quality, vegetation coverage, illumination, and background content. To improve dataset consistency, a preprocessing pipeline was developed consisting of vegetation filtering, blur detection, image normalization, and data augmentation.

3.2.1. Vegetation Filtering

Since the objective was to classify herbal plants, images containing insufficient vegetation were removed. Vegetation regions were identified using color thresholding in the HSV color space. Green pixels corresponding to plant regions were detected and their proportion within the image was computed.

Images containing less than a predefined vegetation coverage threshold were discarded. This step reduced the inclusion of frames dominated by background objects and improved the relevance of the training data.

3.2.2. Blur Detection

Video frames often contain motion blur caused by camera movement or plant motion. To eliminate low-quality samples, image sharpness was evaluated using the Variance of Laplacian method.

The Laplacian operator measures high-frequency image content associated with edges and texture. Images with

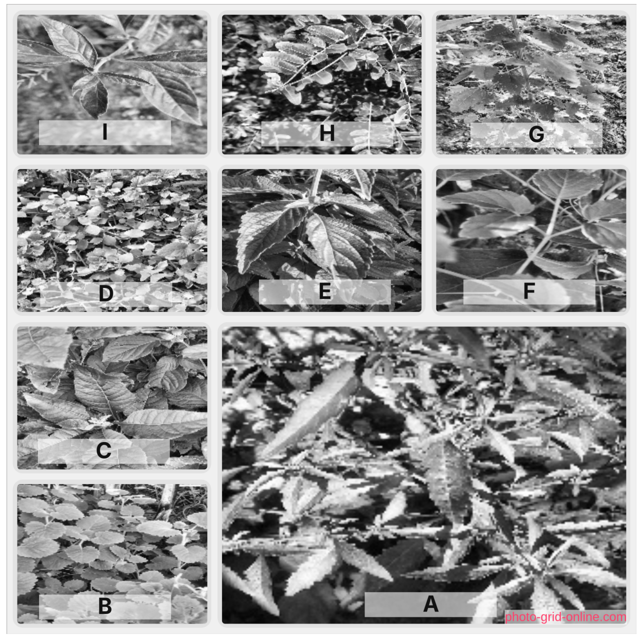

Figure 1. Examples of herbal plant images from different classes

variance values below a predefined threshold were considered blurry and excluded from the dataset.

This filtering process improved the quality of the extracted visual features and reduced noise during model training.

3.2.3. Image Normalization

All retained images were resized to a uniform resolution of 224×224 pixels. For deep learning experiments, pixel intensities were normalized to the range $[0, 1]$ to facilitate stable optimization during training.

3.2.4. Data Augmentation

To increase dataset diversity and reduce overfitting, data augmentation was applied during training. Each image was subjected to random transformations including:

- Horizontal flipping
- Rotation between -20° and 20°
- Brightness adjustment
- Gaussian blurring

These transformations simulate real-world variations in plant appearance and improve model generalization.

3.3. ORB Feature Extraction and Bag-of-Visual-Words Representation

The traditional machine learning pipeline employed Oriented FAST and Rotated BRIEF (ORB) descriptors to capture local image features. ORB was selected because it provides computational efficiency while maintaining robustness to rotation and scale variations.

For each image, up to 1000 ORB keypoints were detected and corresponding descriptors were extracted. The descriptors collected from all training images were combined to form a feature pool containing 232,788 local descriptors.

To convert variable-length descriptor sets into fixed-length feature vectors, a Bag-of-Visual-Words (BoVW) representation was adopted. MiniBatch K-Means clustering was applied to the descriptor pool to construct a visual vocabulary consisting of 500 visual words.

Each image was then represented as a normalized histogram describing the frequency of occurrence of each visual word.

3.4. Traditional Machine Learning Classifiers

The BoVW feature vectors were used to train four traditional machine learning classifiers:

- Support Vector Machine (SVM)
- Random Forest
- Logistic Regression
- Gradient Boosting

Model selection was performed using the validation set. The classifier achieving the highest validation accuracy was selected for final evaluation on the test set.

Hyperparameter optimization for the SVM model was conducted using GridSearchCV with five-fold cross-validation. The search explored different values of the regularization parameter C, kernel types, and kernel scaling parameters.

3.5. Convolutional Neural Network

To investigate feature learning directly from image pixels, a custom Convolutional Neural Network (CNN) was implemented as shown in Fig. 2.

The network architecture consisted of three convolutional blocks followed by fully connected layers:

- Conv(32) + MaxPooling
- Conv(64) + MaxPooling
- Conv(128) + MaxPooling
- Fully Connected Layer (128 neurons)
- Softmax Output Layer (14 classes)

Rectified Linear Unit (ReLU) activation functions were used throughout the network. The model was trained using the Adam optimizer and categorical cross-entropy loss for 10 epochs with a batch size of 32.

3.6. Transfer Learning with MobileNetV2

To overcome the limitations of training deep networks on a relatively small dataset, as illustreated in Fig. 3, transfer learning was employed using MobileNetV2 pre-trained on the ImageNet dataset.

The convolutional backbone of MobileNetV2 was initialized with pre-trained weights and frozen during training. A custom classification head consisting of a Global Average Pooling layer, Dropout layer, and Softmax output layer was added to adapt the model to the herbal plant classification task.

Early stopping was employed to prevent overfitting by monitoring validation loss during training. The transfer learning model was trained for up to 20 epochs and evaluated on the held-out test set.

4. Experimental Results and Discussion

4.1. Experimental Setup

All experiments were conducted using Python and standard machine learning and deep learning libraries. The dataset was divided into training, validation, and testing sets using a stratified 70:15:15 split to ensure balanced class distributions across all subsets.

Three classification approaches were evaluated:

1. ORB feature extraction with Bag-of-Visual-Words (BoVW) representation and traditional machine learning classifiers.
2. A custom Convolutional Neural Network (CNN) trained from scratch.
3. A MobileNetV2 transfer learning model initialized with ImageNet pre-trained weights.

Performance was measured using classification accuracy on both validation and test datasets. The validation set was used for model selection and hyperparameter tuning, while the test set was reserved for final performance evaluation.

4.2. Dataset Summary

After preprocessing and quality filtering, the final dataset contained 425 images distributed across 14 herbal plant classes. The preprocessing pipeline removed low-quality images affected by excessive blur and insufficient vegetation coverage.

The dataset as shown in Tab. 1 presents a challenging classification problem due to variations in lighting conditions, viewing angles, background clutter, and intra-class appearance differences.

4.3. Performance of Traditional Machine Learning Models

The first set of experiments evaluated traditional machine learning algorithms using ORB descriptors and Bag-of-Visual-Words representations.

Table 2 summarizes the validation performance of the evaluated classifiers.

Among the evaluated models, the Support Vector Machine (SVM) achieved the highest validation accuracy of 47.92%, outperforming Random Forest, Logistic Regression, and Gradient Boosting classifiers. Consequently, the SVM model was selected for final testing.

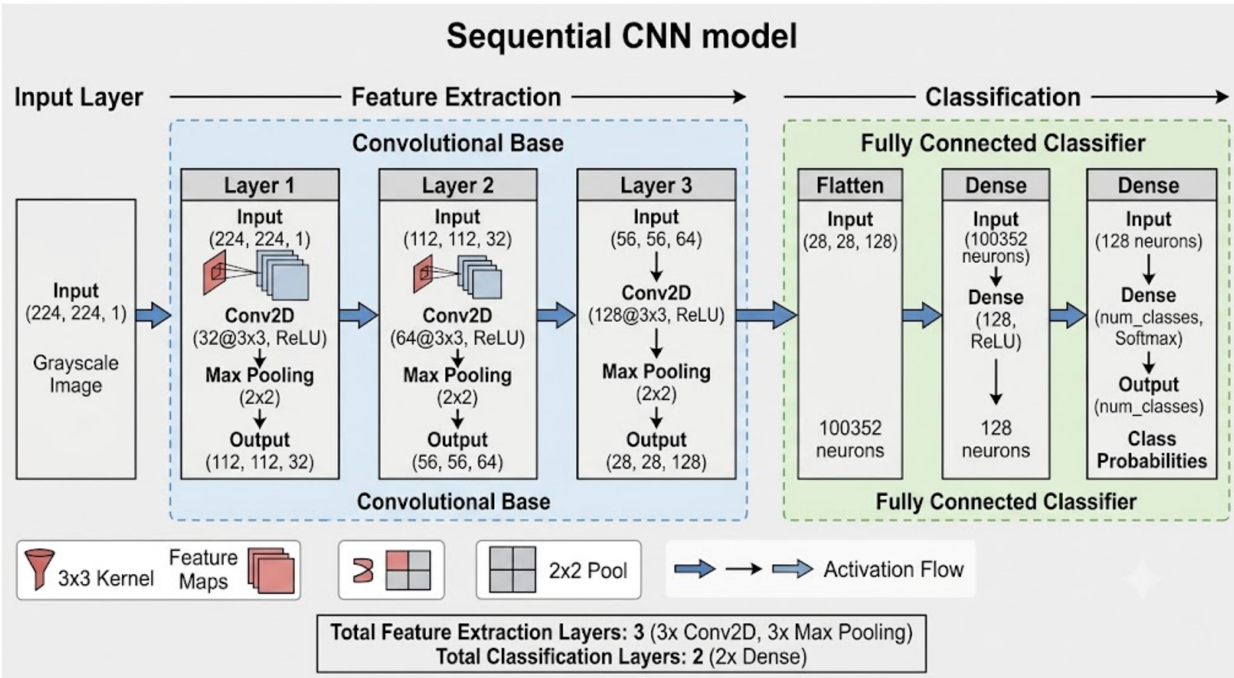

Figure 2. Architecture of the custom CNN model

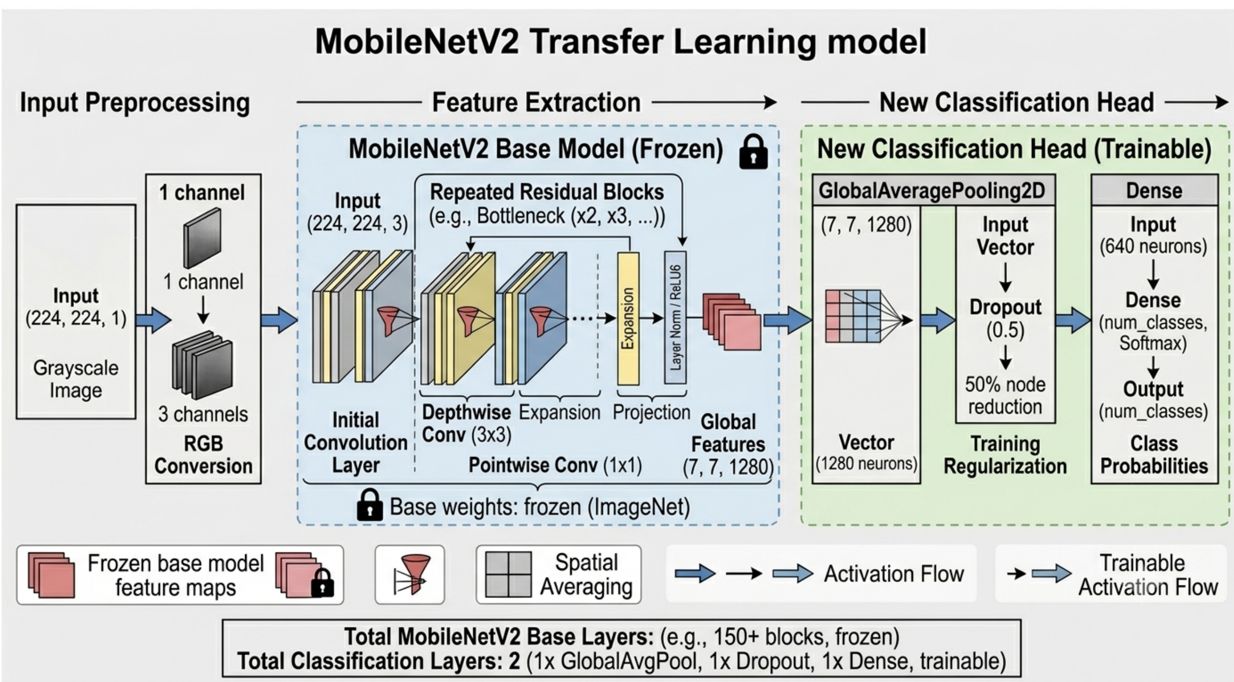

Figure 3. MobileNetV2 transfer learning architecture

When evaluated on the unseen test set, the ORB-BoVW-SVM pipeline achieved a test accuracy of 36.46%. Although the model successfully captured local texture and shape information, its performance was limited by the hand-crafted nature of the extracted features. The Bag-of-Visual-Words representation also discards spatial relationships between image regions, which can be important for distinguishing visually similar plant species.

Table 2. Traditional classifier validation results on the preprocessed plant dataset.

Model	Validation Accuracy (%)
SVM	**47.92**
Logistic Regression	45.83
Random Forest	31.77
Gradient Boosting	28.65

4.4. Performance of the Custom CNN

A custom Convolutional Neural Network was trained from scratch using the herbal plant dataset. Despite its ability to learn hierarchical feature representations directly from image data, the model achieved a test accuracy of only 22.40%.

As shown in Fig. 4, the relatively poor performance can largely be attributed to the limited size of the training dataset. Deep neural networks typically require large numbers of labeled examples to learn robust and generalizable feature representations. With only 425 images distributed across 14 classes, the network likely struggled to learn discriminative features while avoiding overfitting.

Furthermore, plant classification represents a fine-grained recognition task where subtle differences between species must be captured. A shallow custom CNN trained from scratch may not possess sufficient representational capacity to learn these distinctions effectively from a small dataset.

4.5. Performance of MobileNetV2 Transfer Learning

The transfer learning approach based on MobileNetV2 produced the best performance among all evaluated methods. The model achieved a test accuracy of 93.23%, substantially outperforming both the traditional machine learning pipeline and the custom CNN.

The strong performance of MobileNetV2 can be attributed to its pre-trained feature extraction capabilities. Since the network was initially trained on the large-scale ImageNet dataset, it had already learned rich visual representations useful for recognizing textures, shapes, edges, and object structures. Fine-tuning these representations for herbal plant classification required significantly less training data than learning features from scratch.

Additionally, MobileNetV2 employs depthwise separable convolutions, allowing efficient feature extraction while maintaining strong classification performance. This architecture proved particularly suitable for the relatively small and diverse dataset used in this study.

Table 3. Final model comparison evaluated on the test set.

Method	Test Accuracy (%)
MobileNetV2	**93.23**
SVC	36.46
Custom CNN	22.40

4.6. Comparative Analysis

Table **??** presents the final comparison of all evaluated approaches.

The results in Tab. 3 demonstrate a clear performance hierarchy among the evaluated methods. Traditional feature-based approaches achieved moderate performance by leveraging local image descriptors and machine learning classifiers. However, handcrafted features were unable to fully capture the complex visual characteristics required for reliable plant classification.

Interestingly, the custom CNN performed worse than the traditional ORB-BoVW-SVM pipeline. This result highlights an important challenge in deep learning: training neural networks from scratch with limited data often leads to poor generalization. While CNNs possess greater representational power than handcrafted features, their effectiveness depends heavily on the availability of sufficient training examples.

The MobileNetV2 transfer learning approach achieved the highest classification accuracy by a large margin. Compared to the ORB-BoVW-SVM pipeline, MobileNetV2 improved test accuracy by approximately 56.77 percentage points. Compared to the custom CNN, the improvement was approximately 70.83 percentage points.

These findings demonstrate that transfer learning provides a highly effective solution for plant recognition tasks where dataset size is limited. The ability to leverage previously learned visual representations allows deep learning models to achieve excellent performance without requiring extensive data collection efforts.

4.7. Limitations

Several limitations should be considered when interpreting the results. First, the dataset contains only 425 images, which may not fully represent the visual diversity of herbal plant species encountered in real-world environments. Second, the dataset was derived from only 21 video sequences, potentially limiting environmental variability.

Additionally, only image-based classification was investigated. Future work could explore object detection, segmentation, multimodal approaches, or larger datasets containing additional plant species and environmental conditions.

Despite these limitations, the results demonstrate the fea-

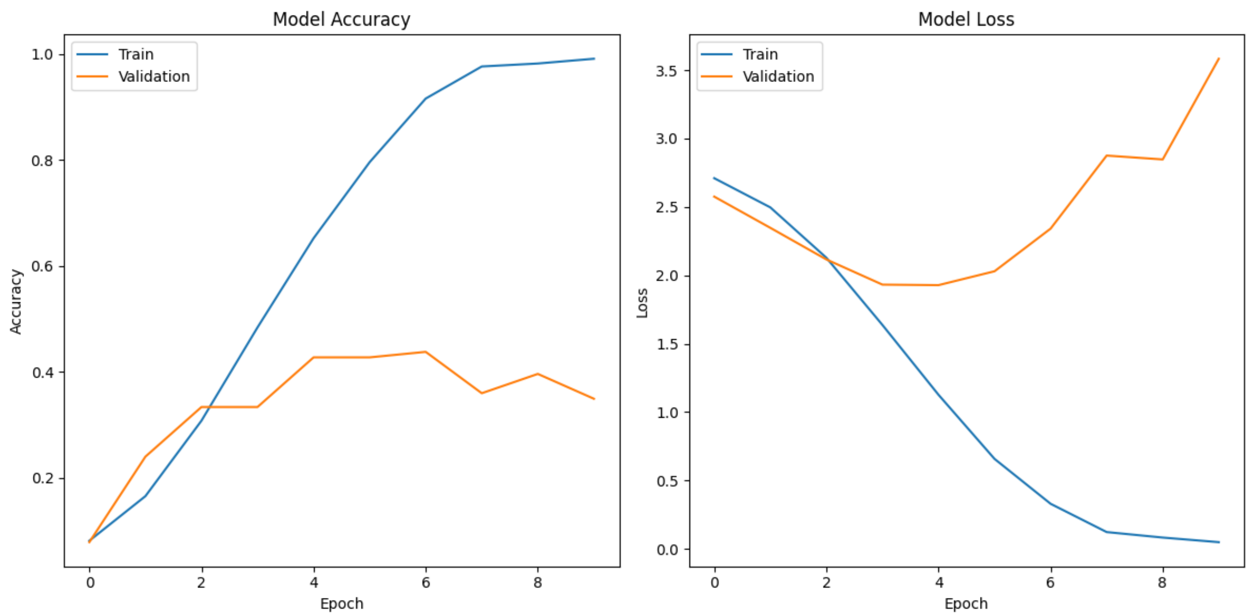

Figure 4. CNN training and validation loss curves

sibility of automated herbal plant classification and highlight the effectiveness of transfer learning for practical plant recognition applications.

5. Conclusion

This paper presented a comparative study of traditional computer vision and deep learning approaches for herbal plant classification using a dataset constructed from 21 video sequences collected under diverse environmental conditions. A preprocessing pipeline incorporating vegetation filtering, blur detection, image normalization, and data augmentation was developed to improve dataset quality and robustness.

Three classification approaches were evaluated: an ORB-based Bag-of-Visual-Words (BoVW) pipeline with traditional machine learning classifiers, a custom Convolutional Neural Network (CNN), and a MobileNetV2 transfer learning model. Experimental results demonstrated substantial differences in performance among the evaluated methods. The traditional ORB-BoVW-SVM approach achieved moderate classification accuracy, while the custom CNN trained from scratch struggled to generalize effectively due to the limited size of the dataset. In contrast, the MobileNetV2 transfer learning model achieved the highest performance with a test accuracy of 93.23%, demonstrating the effectiveness of leveraging pre-trained feature representations for plant recognition tasks.

The results indicate that transfer learning is a practical and highly effective solution for image classification problems involving limited training data. By utilizing knowledge learned from large-scale datasets, transfer learning significantly improves classification accuracy while reducing the amount of task-specific data required for training.

Future work could focus on expanding the dataset with additional plant species and environmental conditions to improve model generalization. Further research may also investigate object detection and instance segmentation techniques to localize plant regions automatically within complex scenes using SIFT[2] or SURF[1]. Additionally, evaluating more advanced transfer learning architectures such as EfficientNet, ConvNeXt, or Vision Transformers could provide further insights into the effectiveness of modern deep learning approaches for herbal plant recognition.

Overall, the findings of this study demonstrate the feasibility of automated herbal plant classification and provide a reproducible framework for future computer vision applications in agriculture, biodiversity conservation, and traditional medicine.

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